

SIX SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

cloud Computing
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define cloud computing.	M1.01	R
2	Name any two public cloud	M1.03	R
3	Processing of multiple tasks simultaneously on multiple processors is called -----	M2.02	R
4	----- is an IT resource that actively filters network traffic to and from the isolated network while controlling its interactions with the internet	M3.01	R
5	List four logical units of data storage	M3.01	R
6	Write two basic Failover systems configurations.	M3.02	R
7	List two primary types of portals that are created with the remote administration system	M3.03	R
8	Mention the two communication services provided by the Amazon web services	M4.01	R
9	Name the type of service provided by Google AppEngine	M4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	List out the goals and benefits of cloud computing	M1.01	R
2	Explain Single-instruction, single-data (SISD) systems.	M2.02	U
3	Define virtualization and List its benefits	M2.04	R
4	Define State management database	M3.02	R

5	Explain about Pay-per-use monitor mechanism and what are the typical monitoring variables.	M3.02	U
6	Explain about SLA management system	M3.03	U
7	List three major agents in Cloud Usage Monitor	M3.01	R
8	Describe Data Store and the type of data that can be stored in it	M4.02	R
9	Identify the type of any three services provided by AWS	M4.01	R
10	Explain the type of storage provided by bucket	M4.01	R

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

III	Explain in detail about Infrastructure as a Service.	M1.02	U
OR			
IV	Describe in detail about Hybrid clouds.	M1.03	U
V	Explain the history of cloud computing in detail.	M1.05	U
OR			
VI	Draw the fundamental block diagram of cloud computing architecture.	M1.04	R
VII	Explain distributed system and characterize its components	M2.03	U
OR			
VIII	List characteristics of virtualized environments?	M2.04	R
IX	Describe the elements of Xen virtualization.	M2.05	U
OR			
X	Differentiate Multiple-instruction, multiple-data (MIMD) systems based on the way PE's are coupled to the main memory.	M2.02	U
XI	Explain the term resource cluster and give the names of common resource cluster types.	M3.02	U
OR			
XII	Define Resource management? List the tasks that are typically automated and implemented through the resource management system	M3.03	R

XIII	Explain the compute services offered by Google AppEngine.	M4.02	U
OR			
XIV	Describe the architecture of SQL Azure with neat diagram	M4.03	U

SIX SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
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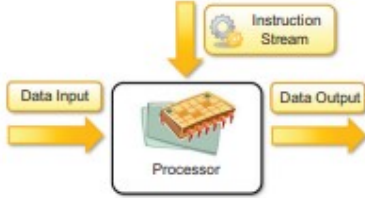
Cloud Computing
Answer Key and Scheme of Evaluation

PART A

1	Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.	1
2	AWS, Google Cloud etc (any two)	0.5*2=1
3	parallel processing	1
4	Virtual Firewall	1
5	Files, Blocks, Datasets, Objects	1
6	Active-active and Active- passive	1
7	Usage and administration portal, self service portal	1
8	Virtual networking and messaging.	1
9	PaaS	1

PART B

1	• Reduced IT costs. Moving to cloud computing may reduce the cost of managing and maintaining your IT systems. ... • Scalability. ... • Business continuity. ... • Collaboration efficiency. ... • Flexibility of work practices. ... • Access to automatic updates. ...	0.5*6=3
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2	An SISD computing system is a uniprocessor machine capable of executing a single instruction, which operates on a single data stream. In SISD, machine instructions are processed sequentially; hence computers adopting this model are popularly called sequential computers. All the instructions and data to be processed have to be stored in primary memory. The speed of the processing element in the SISD model is limited by the rate at which the computer can transfer information internally. Dominant representative SISD systems are IBM PC, Macintosh, and workstations.  SISD Explanation -2 Figure-1	2+1=3
3	It is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources". • Increased performance and computing capacity • Underutilized hardware and software resources • Lack of space • Greening initiatives • Rise of administrative costs Virtualization Definition -0.5 Benefits -2.5	0.5+ (0.5*5) =3
4	State management database (state management database) is a storage device used to temporarily persist the state data of a software program. State management databases are usually used by cloud services, especially services that involve long-running activities.	3
5	The pay-per-use monitor mechanism measures cloud-based IT resource usage in accordance with pre-defined pricing parameters, and generates usage logs to calculate fees. The data collected by the pay-per-use monitor is processed by a billing management system that calculates payment fees. The data collected by the pay-per-use monitor is processed by a billing management system that calculates the payment fees. Typical monitoring variables: (1) Number of request/response messages (2) The amount of data transferred (3) Bandwidth consumption Display the pay-per-use monitor that implements the resource agent to determine the life cycle of the virtual server	Definition - 2 Monitoring variables -1

6	A Service Level Agreement (SLA) is the bond for performance negotiated between the cloud services provider and the client. Earlier, in cloud computing all Service Level Agreements were negotiated between a client and the service consumer. Nowadays, with the initiation of large utility-like cloud computing providers, most Service Level Agreements are standardized until a client becomes a large consumer of cloud services. Service level agreements are also defined at different levels which are mentioned below: • Customer-based SLA • Service-based SLA • Multilevel SLA	SLA Definition - 2 Different levels -1
7	• Monitoring Agent • Resource Agent • Polling Agent	1*3=3
8	DataStore is a service that allows developers to store semistructured data. The service is designed to scale and optimized to quickly access data. DataStore can be considered as a large object data- base in which to store objects that can be retrieved by a specified key. Both the type of the key and the structure of the object can vary. • Paper files • Simple files like a spreadsheet • File systems • Email storage systems (both server and client systems) • Databases	Data Store Definition - 2 Type of data stored - 1
9	Amazon Web Services (AWS) is a cloud computing platform with functionalities such as database storage, delivery of content, and secure IT infrastructure for companies, among others. It is known for its on-demand services namely Elastic Compute Cloud (EC2) and Simple Storage Service (S3). IaaS, PaaS, SaaS	AWS services -2 Type of cloud service -1
10	Buckets are the basic containers that hold your data. Everything that you store in Cloud Storage must be contained in a bucket. You can use buckets to organize your data and control access to your data, but unlike directories and folders, you cannot nest buckets. While there is no limit to the number of buckets you can have in a project or location, there are limits to the rate you can create or delete buckets. When you create a bucket, you give it a globally-unique name and a geographic location where the bucket and its contents are stored. The name and location of the bucket cannot be changed after creation, though you can delete and re-create the bucket to achieve a similar result. There are also optional bucket settings that you can configure during bucket creation and change later.	3

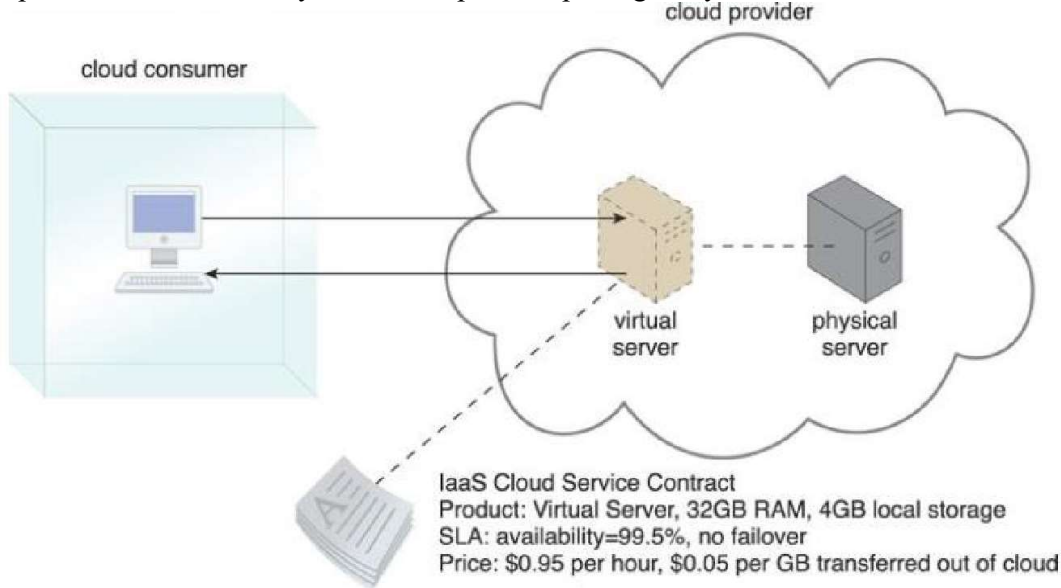
PART C

The IaaS circulation model implies an independent IT climate contained of foundation driven IT assets which will be recovered and accomplished by means of cloud administration based interfaces and devices.

This climate can incorporate equipment, organize, network, working frameworks, and other "crude" IT assets. In distinction to conventional facilitating or redistributing environmental factors, with IaaS, IT assets are normally virtualized and pressed into wraps that compress in advance runtime climbing and customization of the framework. the broadly useful of an IaaS domain is to flexibly cloud customers with an elevated level of control and responsibility over its design and use.

The IT assets gave by IaaS are by and large not pre-arranged, setting the official obligation straightforwardly upon the cloud shopper. This model is consequently utilized by cloud buyers that need a significant level of command over the cloud-based condition they will make. Here and there cloud suppliers will contract IaaS contributions from other cloud suppliers in order to scale their own cloud surroundings. the sorts and makes of the IT assets gave by IaaS items offered by various cloud suppliers can change. IT assets accessible through IaaS conditions are for the most part offered as newly instated virtual occurrences. A focal and first IT asset inside a run of the mill IaaS condition is that the virtual server. Virtual servers are rented by indicating server equipment necessities, similar to processor limit, memory, and local space for putting away

III



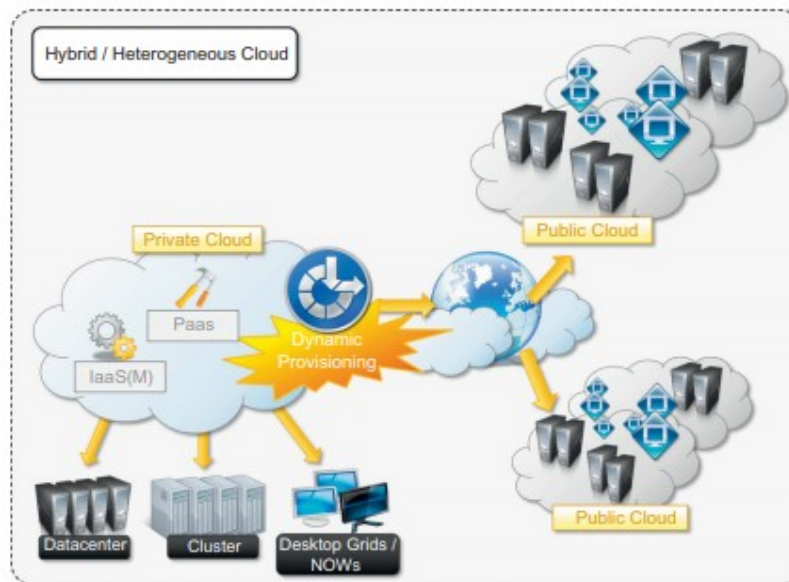
IaaS
Definition -
2
Working -
3
Diagram -
2

IV

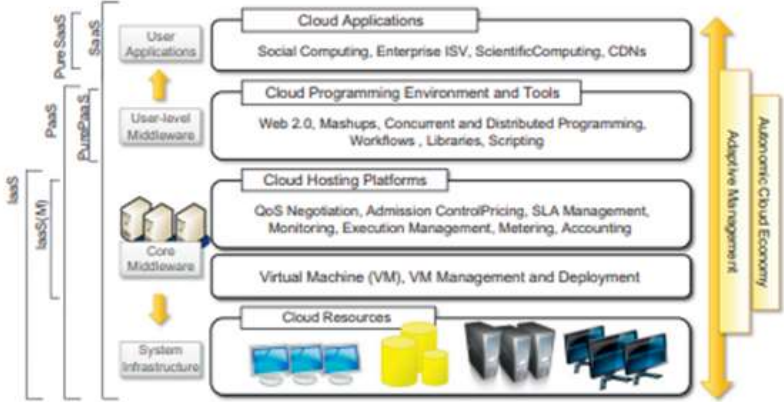
Hybrid clouds allow enterprises to exploit existing IT infrastructures, maintain sensitive information within the premises, and naturally grow and shrink by provisioning external resources and releasing them when they're no longer needed. Security concerns are then only limited to the public portion of the cloud that can be used to perform operations with less stringent constraints but that are still part of the system workload.

It is a heterogeneous distributed system resulting from a private cloud that integrates additional services or resources from one or more public clouds. For this reason they are also called heterogeneous clouds. As depicted in the diagram, dynamic provisioning is a fundamental component in this scenario. Hybrid clouds address scalability issues by leveraging external resources for exceeding capacity demand. These resources or services are temporarily leased for the time required and then released. This practice is also known as cloud bursting.

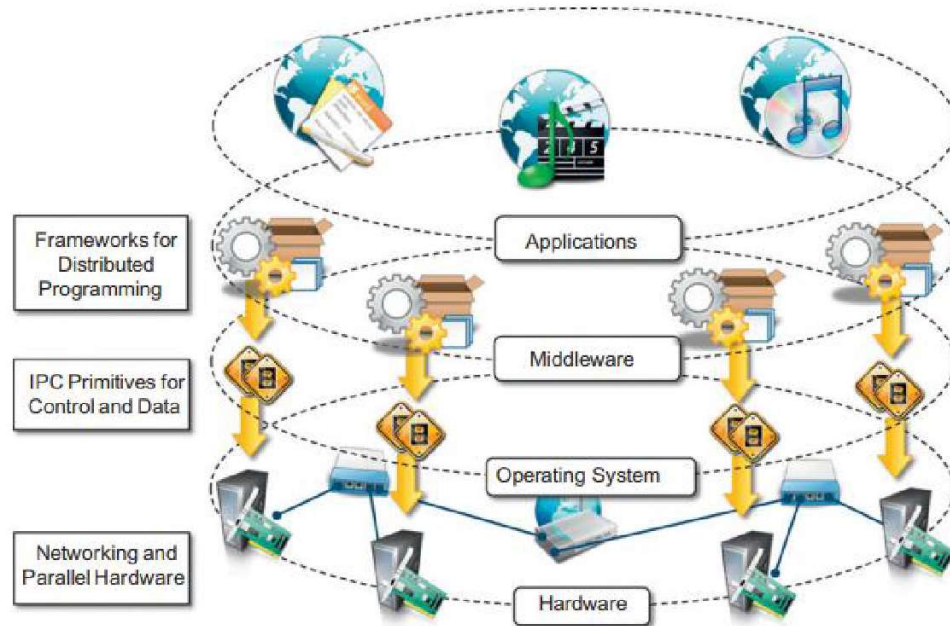
Whereas the concept of hybrid cloud is general, it mostly applies to IT infrastructure rather than software services. Service-oriented computing already introduces the concept of integration of paid software services with existing application deployed in the private premises. In an IaaS scenario, dynamic provisioning refers to the ability to acquire on demand virtual machines in order to increase the capability of the resulting distributed system and then release them. Infrastructure management software and PaaS solutions are the building blocks for deploying and managing hybrid clouds. In particular, with respect to private clouds, dynamic provisioning introduces a more complex scheduling algorithm and policies, the goal of which is also to optimize the budget spent to rent public resources.



Hybrid
Cloud
Definition
-2
Architectu
re -2
Explanatio
n -3

V	Before emerging the cloud computing, there was Client/Server computing which is basically a centralized storage in which all the software applications, all the data and all the controls are resided on the server side. If a single user wants to access specific data or run a program, he/she need to connect to the server and then gain appropriate access, and then he/she can do his/her business. Then after, distributed computing came into picture, where all the computers are networked together and share their resources when needed. On the basis of above computing, there was emerged of cloud computing concepts that later implemented. At around in 1961, John MacCharty suggested in a speech at MIT that computing can be sold like a utility, just like a water or electricity. It was a brilliant idea, but like all brilliant ideas, it was ahead if its time, as for the next few decades, despite interest in the model, the technology simply was not ready for it. But of course time has passed and the technology caught that idea and after few years we mentioned that: <i>In 1999, Salesforce.com started delivering of applications to users using a simple website.</i> The applications were delivered to enterprises over the Internet, and this way the dream of computing sold as utility were true. <i>In 2002, Amazon started Amazon Web Services, providing services like storage, computation and even human intelligence. However, only starting with the launch of the Elastic Compute Cloud in 2006 a truly commercial service open to everybody existed.</i> <i>In 2009, Google Apps also started to provide cloud computing enterprise applications.</i> Of course, all the big players are present in the cloud computing evolution, some were earlier, some were later. <i>In 2009, Microsoft launched Windows Azure, and companies like Oracle and HP have all joined the game. This proves that today, cloud computing has become mainstream.</i>	7
VI	Architecture  Diagram - 3 Marks Explanation - 4 Marks	7

A distributed system is one in which components located at networked computers communicate and coordinate their actions only by passing messages. Nearly all large computing systems are distributed now. Systems are distributed. The distributed system is "a set of independent machines that present to the user as one coherent system." Information processing is distributed on several machines instead of being confined to a single computer.



At the lowest level the physical infrastructure is computer and network hardware, which is explicitly supervised by the operating system that provides the essential interprocess communication (IPC) services, the scheduling and management of the process, as well as the management of resources in file systems and local systems.

The implementation of well-known operating system principles and many more on a hardware and network level allows heterogeneous components and their structure to be integrated easily into a consistent, unified framework. For instance, connectivity in the network among various devices is governed by standards, allowing for smooth interaction. The middleware layer utilizes such services to develop and deploy distributed applications in a consistent environment. When using the services provided by the operating system, middleware creates its own protocols, data formats and programming language or frameworks to create distributed apps.

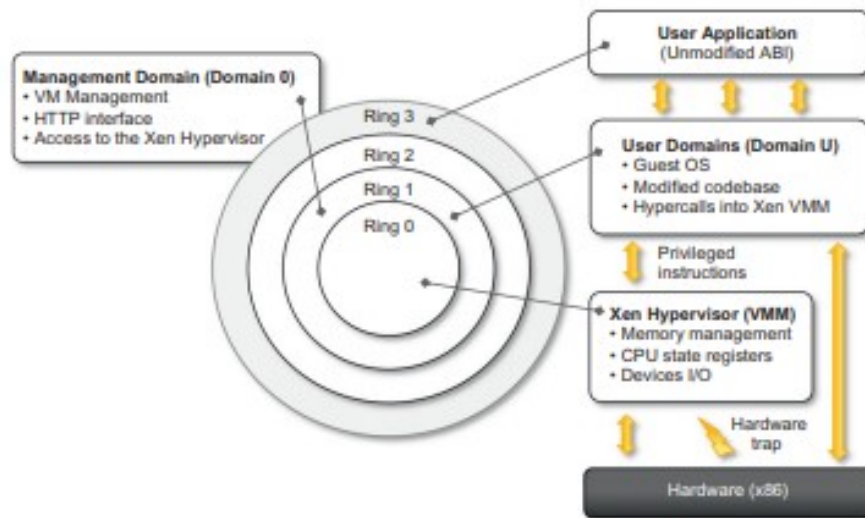
The applications and services designed and built for middleware use reflect the upper part of the distributed system stack. These can be used for many reasons. Sometimes they can view their functionality through a web browser in the form of graphical interfaces (GUIs).

Distributed System
Definition -2
Architecture -2
Explanation -3

VII

VIII	Increased Security – The ability to fully transparently govern the execution of a guest program creates new opportunities for providing a safe, controlled execution environment. All guest programs operate usually against the virtual machine, translating them and using them for host program. Execution Managed – In particular, the most important features are sharing, aggregation, emulation and isolation. Sharing – Virtualization makes it possible to create a separate computing environment in the same host. This common function reduces the amount of active servers and reduces energy consumption. Aggregation – The physical resource can not only be shared between several guests, but virtualization also enables aggregation. A group of individual hosts can be linked and represented as a single virtual host. This functionality is implemented using the Cluster Management Software, which uses and represents the physical resources of a uniform group of machines. Emulation – In the virtualization layer, which is essentially a program, guest programs are executed within an environment. An entirely different environment can also be emulated with regard to the host, so that guest programs that require certain features not present in the physical host can be carried out. Isolation – Virtualization allows guests to provide an entirely separate environment in that they are executed — if they are operating systems, applications or other entities. The guest program operates through an abstraction layer that offers access to the underlying resources. The virtual machine is able to filter the guest's activities and prevent dangerous operations against the host. Portability – In the case of a hardware virtualization, the guest is packed in a virtual image which can be moved and executed safely on various virtual machines in many instances. With the virtualization of the programming level, as carried out in JVM or in .NET runtime, the binary code of the application components (jars or assemblies) may work on the respective virtual machine without recompilation.	7*1=7
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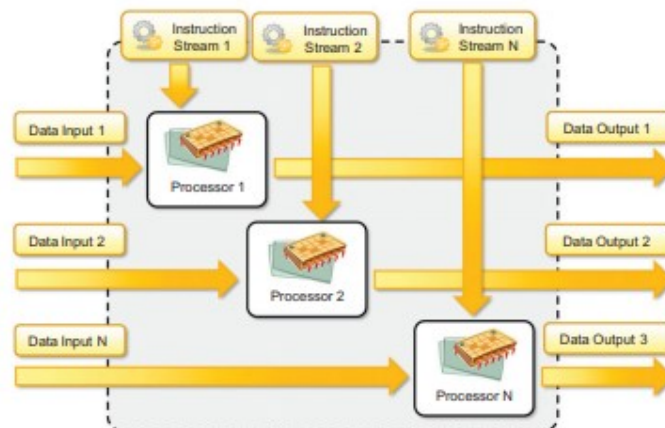
Xen is an open-source initiative implementing a virtualization platform based on paravirtualization. Initially developed by a group of researchers at the University of Cambridge in the United Kingdom, Xen now has a large open-source community backing it. Citrix also offers it as a commercial solution, XenSource. Xen-based technology is used for either desktop virtualization or server virtualization, and recently it has also been used to provide cloud computing solutions by means of Xen Cloud Platform (XCP). At the basis of all these solutions is the Xen Hypervisor, which constitutes the core technology of Xen. Recently Xen has been advanced to support full virtualization using hardware-assisted virtualization. Xen is the most popular implementation of paravirtualization, which, in contrast with full virtualization, allows high-performance execution of guest operating systems.



A Xen-based system is managed by the Xen hypervisor, which runs in the highest privileged mode and controls the access of guest operating system to the underlying hardware. Guest operating systems are executed within domains, which represent virtual machine instances. Moreover, specific control software, which has privileged access to the host and controls all the other guest operating systems, is executed in a special domain called Domain 0. This is the first one that is loaded once the virtual machine manager has completely booted, and it hosts a HyperText Transfer Protocol (HTTP) server that serves requests for virtual machine creation, configuration, and termination. This component constitutes the embryonic version of a distributed virtual machine manager, which is an essential component of cloud computing systems providing Infrastructure-as-a-Service (IaaS) solutions.

Xen
Definition
-2
Architectu
re -2
Explatio
n -3

An MIMD computing system is a multiprocessor machine capable of executing multiple instructions on multiple data sets. Each PE in the MIMD model has separate instruction and data streams; hence machines built using this model are well suited to any kind of application. Unlike SIMD and MISD machines, PEs in MIMD machines work asynchronously.



MIMD machines are broadly categorized into shared-memory MIMD and distributed-memory MIMD based on the way PEs are coupled to the main memory.

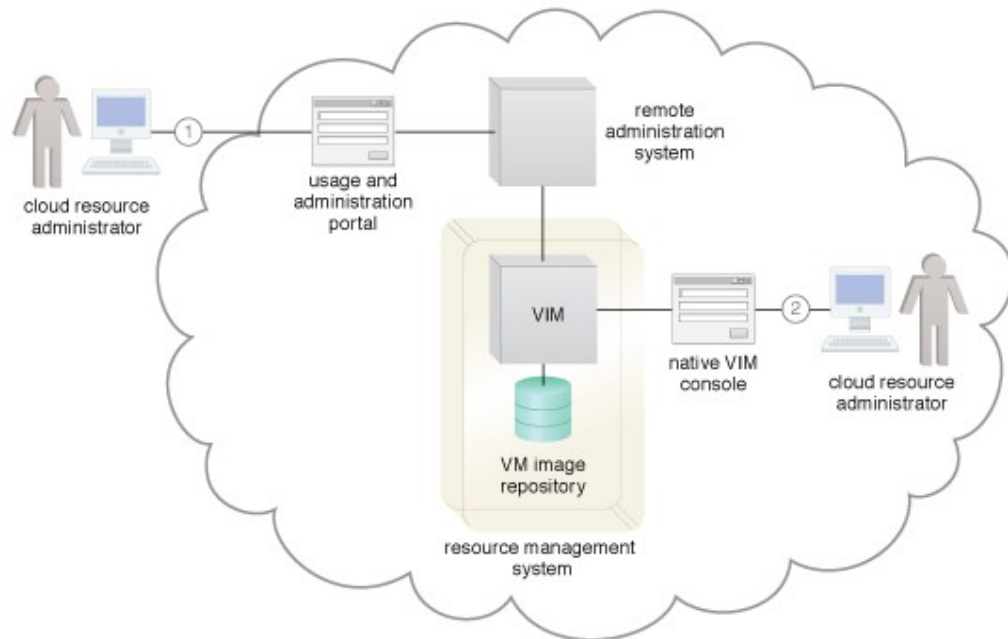
Shared memory MIMD machines- In the shared memory MIMD model, all the PEs are connected to a single global memory and they all have access to it (see Figure 2.6). Systems based on this model are also called tightly coupled multiprocessor systems. The communication between PEs in this model takes place through the shared memory; modification of the data stored in the global memory by one PE is visible to all other PEs. Dominant representative shared memory MIMD systems are Silicon Graphics machines and Sun/IBM's SMP (Symmetric Multi-Processing).

Distributed memory MIMD- machines In the distributed memory MIMD model, all PEs have a local memory. Systems based on this model are also called loosely coupled multiprocessor systems. The communication between PEs in this model takes place through the interconnection network (the interprocess communication channel, or IPC). The network connecting PEs can be configured to tree, mesh, cube, and so on. Each PE operates asynchronously, and if communication/synchronization among tasks is necessary, they can do so by exchanging messages between them.

MIMD
Definition
-2
Diagram -
1
Shared
Memory
MIMD -2
Distribute
d Memory
MIMD -2

XI	Resource cluster (resource cluster) The resource cluster mechanism is to group multiple IT resource instances into a group so that they can operate like an IT resource. This enhances the combined computing power, load balancing capability and availability of IT resources. The resource cluster mechanism increase the combined computing capacity, load balancing, and availability of the clustered IT resources. Common resource cluster types: (1) Server cluster -physical or virtual servers form a cluster to improve performance and availability. (2) Database cluster - this kind of highly available resource cluster is used to improve data availability, has synchronization characteristics, and can maintain the consistency of data stored on various storage devices used in the cluster. (3) Large dataset cluster -to achieve the partition and distribution of data, so that the target data set can be effectively divided into regions, without the need to destroy the integrity of the data and the accuracy of calculations. Two basic types of resource clusters (1) Load balanced cluster—The specialty of this resource cluster is to distribute workload among cluster nodes. It usually implements a load balancer mechanism. (2) HA cluster (HA cluster)-It implements a failover system mechanism, monitors failure conditions, and automatically redirects workloads away from the failed node.	Resource Cluster Definition -2 Server cluster -1 Database cluster -1 Large dataset cluster -1 Load balanced cluster -1 HA cluster -1
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The resource management system mechanism helps coordinate IT resources in response to management actions performed by both cloud consumers and cloud providers. Core to this system is the virtual infrastructure manager (VIM) that coordinates the server hardware so that virtual server instances can be created from the most expedient underlying physical server. A VIM is a commercial product that can be used to manage a range of virtual IT resources across multiple physical servers. For example, a VIM can create and manage multiple instances of a hypervisor across different physical servers or allocate a virtual server on one physical server to another (or to a resource pool).

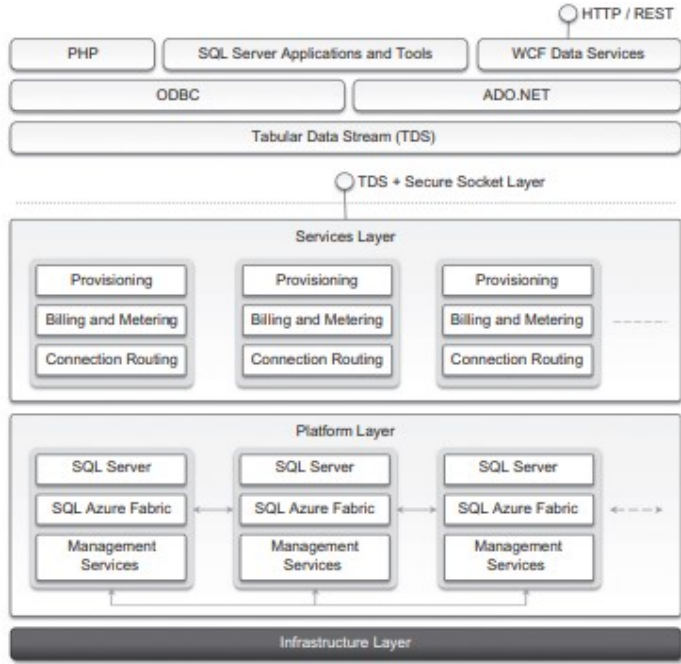


Tasks that are typically automated and implemented through the resource management system include:

- managing virtual IT resource templates that are used to create pre-built instances, such as virtual server images
- allocating and releasing virtual IT resources into the available physical infrastructure in response to the starting, pausing, resuming, and termination of virtual IT resource instances
- coordinating IT resources in relation to the involvement of other mechanisms, such as resource replication, load balancer, and failover system
- enforcing usage and security policies throughout the lifecycle of cloud service instances
- monitoring operational conditions of IT resources

Resource Management
Definition
-2
Figure -2
Task -3

XIII	AppEngine offers additional services such as Task Queues and Cron Jobs that simplify the execution of computations that are off-bandwidth or those that cannot be performed within the timeframe of the Web request. Task queues Task Queues allow applications to submit a task for a later execution. This service is particularly useful for long computations that cannot be completed within the maximum response time of a request handler. The service allows users to have up to 10 queues that can execute tasks at a configurable rate. In fact, a task is defined by a Web request to a given URL, and the queue invokes the request handler by passing the payload as part of the Web request to the handler. It is the responsibility of the request handler to perform the “task execution,” which is seen from the queue as a simple Web request. The queue is designed to reexecute the task in case of failure in order to avoid transient failures preventing the task from a successful completion. Cron jobs Sometimes the length of computation might not be the primary reason that an operation is not performed within the scope of the Web request. It might be possible that the required operation needs to be performed at a specific time of the day, which does not coincide with the time of the Web request. In this case, it is possible to schedule the required operation at the desired time by using the Cron Jobs service. This service operates similarly to Task Queues but invokes the request handler specified in the task at a given time and does not reexecute the task in case of failure. This behavior can be useful to implement maintenance operations or send periodic notifications.	Task queues -4 Cron jobs -3
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XIV	SQL Azure is a relational database service hosted on Windows Azure and built on the SQL Server technologies. The service extends the capabilities of SQL Server to the cloud and provides developers with a scalable, highly available, and fault-tolerant relational database. SQL Azure is accessible from either the Windows Azure Cloud or any other location that has access to the Aazure Cloud. It is fully compatible with the interface exposed by SQL Server, so applications built for SQL Server can transparently migrate to SQL Azure. Moreover, the service is fully manageable using REST APIs, allowing developers to control databases deployed in the Azure Cloud as well as the firewall rules set up for their accessibility  The diagram illustrates the SQL Azure architecture layers. At the top, client applications (PHP, SQL Server Applications and Tools, WCF Data Services) connect via HTTP/REST. These connect to ODBC and ADO.NET drivers, which then use the Tabular Data Stream (TDS) protocol. A 'TDS + Secure Socket Layer' connection leads to the 'Services Layer', which contains Provisioning, Billing and Metering, and Connection Routing services. Below this is the 'Platform Layer', which includes SQL Server instances, SQL Azure Fabric, and Management Services. The entire stack sits on the 'Infrastructure Layer'. Access to SQL Azure is based on the Tabular Data Stream (TDS) protocol, which is the communication protocol underlying all the different interfaces used by applications to connect to a SQL Server-based installation such as ODBC and ADO.NET. On the SQL Azure side, access to data is mediated by the service layer, which provides provisioning, billing, and connection-routing services. These services are logically part of server instances, which are managed by SQL Azure Fabric. This is the distributed database middleware that constitutes the infrastructure of SQL Azure and that is deployed on Microsoft data centers	SQL Azure Definition -2 Architectu re -2 Explanatio n -3
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